

WHOM TO SCREEN?

INDIVIDUALS WITH FOLLOWING RISK FACTORS

TARGET POPULATION: AGE > 18 YEARS



DIABETES



HYPERTENSION



CARDIOVASCULAR
DISEASE



OBESE



SMOKING



FAMILY HISTORY



HISTORY OF ACUTE
KIDNEY INJURY



*USE OF NEPHROTOXIC
DRUGS



CONNECTIVE TISSUE
DISEASE LIKE SLE



AGE > 60



KIDNEY
STONES

*NSAIDS: DICLOFENAC, ACECLOFENAC, NIMESULIDE, IBUPROFEN,
ETORICOXIB, ETC

AMINOGLYCOSIDES: GENTAMICIN, AMIKACIN, TOBRAMYCIN
SOME HERBAL MEDICINE

CLINICAL PRESENTATION

PERSONS MAY NOTICE SYMPTOMS ONLY IN ADVANCED STAGES
AND SYMPTOMS ARE NONSPECIFIC



SHORTNESS OF
BREATH



FREQUENT
URINATION
DURING NIGHT



HIGH BLOOD
PRESSURE



NAUSEA



FATIGUE



FOAMY/FROTHY
OR RED URINE



SWELLING IN
LEGS



DECREASED
URINATION

CRITERIA FOR REFERRAL

URGENT REFERRAL

- Sudden decline in eGFR
- Hyperkalemia
- Resistant hypertension
- Shortness of breath

EARLY REFERRAL

- eGFR < 30 ml/min/1.73m²
- First detected CKD
- Persistent dipstick detected proteinuria
- Blood in the urine
- Recurring kidney stones
- Hereditary kidney diseases e.g. ADPKD

Note: Advise vaccination for Hepatitis B and Tetanus
(Pneumococcal vaccine and Influenza vaccine if available)

SICK DAY RULE

- STOP METFORMIN
- STOP/DECREASE ENALAPRIL/LOSARTAN
- STOP EMPAGLIFLOZIN/DAPAGLIFLOZIN

Calculate eGFR
Using the QR. Code



CONTACT US

For any clinical queries / If anything in
the document is difficult to understand



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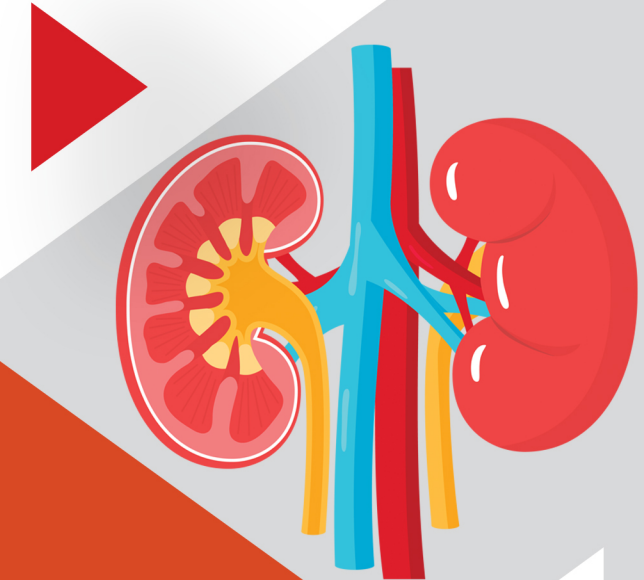
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CHRONIC KIDNEY DISEASE (CKD) IDENTIFICATION AND MANAGEMENT FOR PRIMARY CARE PROVIDERS

USE THIS GUIDE TO

- Identify patients with CKD
- Screen for CKD among patient at risk
- Manage and refer patients with CKD

Chronic kidney disease is defined as reduced kidney function
and / or structural damage of kidney for more than 3 months.
Approximately one in five adults with hypertension and one
in three adults with diabetes have CKD

INVESTIGATIONS

Serum creatinine, Random Blood Sugar, Urinary dipstick for albumin, Albumin creatinine ratio*, 24 hour urinary protein*, Lipid profile*, Ultrasonography* (* उपलब्ध अणुमा)

3 TESTS FOR KIDNEY HEALTH

TEST
1

B.P. CHECK

Blood Pressure constantly maintained below BP goal



URINE TEST

Albumin/Creatinine Ratio(ACR) to check for albuminuria

Urine dipstick test for albumin and RBCs(blood)

TEST
2

TEST
3

BLOOD TEST

eGFR calculated from serum creatinine



TREATMENT TARGET

Blood pressure target <130/80 mmHg
HbA1c target: 6.5 - 7%

Note: The target may be higher (<8%) in patients with frailty and CKD stage 5

LDL target <100 mg/dL

Hemoglobin target : 10-11gm/ dL

“CKD is common, preventable and easy to treat if detected early!”

DIAGNOSIS

eGFR LESS THAN 60 ML/MIN/1.73M² FOR MORE THAN 3 MONTHS WITH OR WITHOUT:

1. Proteinuria (Urinary ACR > 30 mg/g or 24 hours UTA > 300 mg) / Persistent 1+ albumin in dipstick test
2. Abnormal kidneys on imaging
3. Hematuria, red cell cast, white cell cast

*eGFR: estimated Glomerular Filtration Rate, ACR: Albumin creatinine ratio, UTA: Urinary total albumin

Calculate eGFR
Using the QR. Code



ASSESSMENT OF STAGE OF KIDNEY DISEASE

	ALBUMINURIA RANGE		
	A1 <30 MG/G	A2 30-300 MG/G	A3 >300 MG/G
eGFR RANGE	G1 ≥90		
	G2 60-89		
	G3A 45-59		
	G3B 30-44		
	G4 15-29		
	G5 <15		
	Low risk (if no other markers of kidney disease, no CKD)	Moderately increased risk	High risk
			Very high risk

Urinary ACR

Ref: KDIGO, Kidney International 2025

MANAGEMENT

1. BLOOD PRESSURE CONTROL

Is blood pressure more than 130/80 mm Hg ?

YES

Is eGFR > 30ml/min/1.73m²

YES

Give Enalapril 2.5 mg BD or
Losartan 50 mg OD and
measure: creatinine &
potassium in 2 weeks

NO

Give Amlodipine 5 mg PO
OD to a maximum dose of
10 mg PO OD and REFER

Refer “KHDC HYPERTENSION LEAFLET”

2. BLOOD SUGAR CONTROL

Refer “KHDC TYPE 2 DIABETES MELLITUS LEAFLET”

3. STOP AND AVOID NEPHROTOXIC DRUGS

Note: Decrease in eGFR and increase in albuminuria (albumin in urine) shows the progression of kidney diseases



NON-PHARMACOLOGICAL ADVICE



Regular Physical
Activity



Reduce
Stress



Low Salt
Consumption



Quit Alcohol
& Tobacco



Consume
Healthy Diet



Weight
Reduction



Adequate
Sleep

Note:

- Adequate fluid intake especially in case with recurrent renal stones and urinary infections
- Minimize high potassium containing foods like fruits, dry fruits, green leafy vegetables in eGFR < 30